

Shay Mordechai

Security Researcher | Vulnerability Research, Reverse Engineering & OS Internals

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Summary

Offensive Security Researcher specializing in vulnerability discovery, Reverse Engineering, and OS Internals. Proven expertise in bypassing anti-analysis mechanisms, auditing complex execution environments (JIT, Compilers), and authoring exploit reports.

Professional Experience

Security Architect | MyLeads AI (Independent) | Jan 2026 – Present

- Red Team Projects:** Executed Web penetration testing via Burp Suite. Developed novel bypass techniques and advanced TTPs to uncover critical API logic flaws and IDORs.
- Secure API Architecture:** Architected encrypted microservice tunnels using Envoy Proxy filters to dynamically redact egress traffic and prevent browser-side data leakage.
- OS Isolation:** Hardened execution environments bypassing virtualization using Rootless Podman and Linux namespaces.

Application Security Intern | Israel Tax Authority | Aug 2024 – Jul 2025

- AppSec & Defense-in-Depth:** Integrated Checkmarx SAST/SCA into 15+ pipelines, training R&D on OWASP Top 10. Managed Threat Modeling and hardened IBM DataPower gateways via strict JSON validation schemas.

Combat Soldier | Intelligence Collection Corps, IDF | Aug 2016 – Apr 2018

- Executed covert reconnaissance operations, deploying stealth observation posts for operational target profiling.

Education

B.Sc. Computer Science | Jerusalem College of Technology | 2021 – 2025 (GPA: 89)

- Coursework: Reverse Engineering (93), Network Analysis (97), InfoSec (90).

Projects & Research (Detailed Write-ups at shay-mordechai.github.io)

- Advanced RE Challenges:** Solved highly selective challenges (ISA/Shabak Level 1). Analyzed multi-stage malware, bypassed TLS Callbacks/anti-debugging, executed Entry Point Hijacking, and extracted hidden PE payloads via custom C++ decryptors.
- AI-Augmented Vuln Research:** Executed pre-deployment red-teaming via semantic fuzzing. Discovered 3 pending Zero-Days (CERT-IL #11265), aiding in secure model hardening.
- Network Exfiltration & C2:** Engineered a CTF platform modeling nation-state tactics, implementing ICMP covert channels, PKI MITM, and steganography.
- V8 JIT Internals:** Published comprehensive reports on browser engine vulnerabilities, utilizing detours, hooking, and runtime code manipulation to achieve CFI bypasses.
- TLS 1.2 Simulator:** Authored a 4,000+ line Python/Scapy framework to simulate full TLS 1.2 handshakes from scratch, including automated SSLKeyLog generation for DPI.

Technical Skills

- Tools:** IDA Pro, x32dbg, WinDbg, Wireshark, Burp Suite, Sysinternals, Checkmarx.
- Languages:** C/C++, x86 Assembly, Python, Bash. Knowledge of JS, Java, C#, Rust, Raku.
- Bug Bounty & CTFs:** Palo Alto Networks & TP-Link PSIRT Acknowledgments (DNS Rebinding bypasses). Experienced in advanced RE environments (crackmes.one, Flare-On, defense agency CTFs).